

Convexification of Atomic-Structure Solution from the Pair Distribution Function — Minimal Chemical Prior and Resolution

Motivation

Determining atomic coordinates from the pair distribution function (PDF) is the central inverse problem of total-scattering crystallography [4]. Mathematically, the PDF inversion is an instance of the *unassigned* Distance Geometry Problem (uDGP) [3]: the data are a multiset (distribution) of interatomic distances without the assignment of each distance to a specific atom pair. Empirically it is *non-convex*: practical solvers need good initial guesses, restraints, and multi-start global optimization. The non-convexity has two separable sources. The first consists of *trivial* symmetries — permutations of identical atoms, rigid motions, and chirality — which one may quotient out by convention. The second is a *residual geometric* non-uniqueness of the distance-to-coordinate map: several non-congruent structures can share the same pair-distance data, and band-limited instrumental resolution erases further information. Chemical knowledge (bond lengths, angles, coordination, steric exclusion) is known in practice to “convexify” the problem by pruning spurious minima. The question we pose is *how much* chemical information, *jointly with the instrument resolution*, suffices to make the residual problem identifiable and convex — equivalently, what is the *minimal* extra information about the structure that the PDF alone lacks?

Setup

Let $X = (x_1, \dots, x_N) \in (\mathbb{R}^3)^N$ be a structure whose atoms carry element labels $e_i \in E$ (a finite set) with known scattering lengths $b_i > 0$. We quotient by rigid motions and write

$$\mathcal{X} = (\mathbb{R}^3)^N / E(3),$$

where $E(3)$ acts by translations and rotations only; reflections are *not* quotiented, so enantiomers are distinct points of $\mathcal{X}$. The (scattering-weighted) *pair-distance measure* of X is

$$\rho_X = \sum_{i < j} w_{ij} \delta_{\|x_i - x_j\|}, \quad w_{ij} = b_i b_j \geq 0,$$

a finite positive measure on $\mathbb{R}_{\geq 0}$. The experimental PDF is a *band-limited, broadened* linear measurement of ρ_X ,

$$F(X) = R_{Q_{\max}, \sigma} \rho_X \in Y,$$

where $R_{Q_{\max}, \sigma} : \mathcal{M}(\mathbb{R}_{\geq 0}) \rightarrow Y$ is the known *resolution operator*. Physically the reduced PDF is $G(r) = \frac{2}{\pi} \int_0^{Q_{\max}} Q [S(Q) - 1] \sin(Qr) dQ$ [4]; $R_{Q_{\max}, \sigma}$ encodes (i) the reciprocal-space window supported on $[0, Q_{\max}]$, whose truncation produces termination ripples and sets the real-space resolution $\Delta r \approx \pi/Q_{\max}$, and (ii) convolution in r with a kernel of width $\sigma \approx \sigma_{\text{res}}(Q_{\max})$ modelling instrumental and thermal broadening. The essential property is that $R_{Q_{\max}, \sigma}$ is a

compact smoothing operator with non-trivial kernel: features of ρ_X below the scale Δr are invisible. Let

$$m_{\text{eff}}(Q_{\text{max}}, \sigma) = \dim \text{span} \{ R_{Q_{\text{max}}, \sigma} \delta_t : t \in [0, R_{\text{max}}] \} \approx \lceil R_{\text{max}} / \Delta r \rceil$$

be the *instrument information capacity*: the number of linearly independent band-limited scalar constraints the measurement can deliver over the relevant distance range $[0, R_{\text{max}}]$ (reduced by the broadening σ). It is finite, monotone in Q_{max} , with $m_{\text{eff}} \rightarrow \infty$ as $Q_{\text{max}} \rightarrow \infty$ (the exact-distance limit, where every distinct distance is separately resolvable) and $m_{\text{eff}} \rightarrow 0$ as $Q_{\text{max}} \rightarrow 0$. For any structure, $\text{rank}(DF_X) \leq \min(d_C, m_{\text{eff}})$, so the data can constrain at most m_{eff} independent degrees of freedom of the prior.

Chemical prior (hybrid formalization). A *chemistry-consistent prior* is a closed subset $C \subset \mathcal{X}$ that is required to be realizable as an *internal-coordinate constraint set*: a bond graph $G_b = (V, E_b)$ with fixed bond lengths and fixed bond angles, a specified set of free dihedral angles (so that the relaxed geometry is a smooth manifold M_C of dimension $d_C = \#\{\text{free dihedrals}\}$), together with inequality constraints encoding steric exclusion (hard-sphere lower bounds $r_{\min}(e_i, e_j) > 0$ on non-bonded distances), element-specific coordination, and hybridization. We require C to satisfy the *chemistry-consistency axioms*: positive steric radii; element-specific admissible bond-length and bond-angle intervals; and coordination bounds. Concrete priors (a Z -matrix, a fixed topology, hard-sphere exclusion) are instances; C is otherwise treated abstractly.

Known facts

1. **Unassigned distance geometry.** The PDF inversion belongs to the unassigned Distance Geometry Problem (uDGP) framework [3], which formalizes reconstruction from a list of distances *without* the assignment of distances to vertex pairs; the same work treats distance-interval uncertainty and vector geometry problems (VGPs) arising in PDF methods for nanostructures. Our problem extends uDGP to *band-limited, scattering-weighted* distance distributions on a chemistry-constrained manifold.
2. **Exact-distance, infinite-resolution limit ($R = \text{id}$).** Injectivity of ρ (equivalently of the Euclidean distance matrix) up to rigid motion is the classical distance-geometry / global-rigidity question. In dimension 3, generic global rigidity requires redundant rigidity and stressability and is subtler than edge-counting; symmetric or special structures (crystals, icosahedra) are generically *not* globally rigid, so discrete “flip” ambiguities persist even with perfect data.
3. **One-dimensional distance-distribution case.** For points on a line or a loop with unassigned *noisy* pairwise distances (the turnpike / beltway problems), the distance distribution is a collection of quadratic functionals of an N -hot density, and projected gradient descent with a spectral initializer converges *locally linearly* to the global optimizer under explicit conditions [1]. This is the one-dimensional realization of “convexification of the distance-distribution inversion” — but only in 1D and without a chemical prior.
4. **Sparsity-driven recoverability.** For integer sets, uniqueness of turnpike reconstruction is open, yet *sparsity* of the integer set yields polynomial-time recovery up to shift and reversal with high probability when the sparsity is $O(n^{1/2-\varepsilon})$ [2]. This is an instance of “extra structural information (here sparsity) makes recovery possible” — the discrete analogue of the chemical prior in our problem.
5. **Irreducible residue.** Chirality cannot be resolved by any pair-distance data: enantiomers have identical ρ_X and hence identical $F(X)$ for every R . Identifiability therefore holds at

best modulo the $\mathbb{Z}_2$ reflection; the fiber cardinality satisfies $K \geq 2$ for chiral structures. This is an information bound, not a measurement-precision issue.

Open problem

For data $y = F(X_\star)$ and the least-squares loss $\mathcal{L}_y(X) = \|F(X) - y\|_Y^2$ restricted to C , consider:

- (P1) *Identifiability.* $F|_C$ is injective modulo chirality: every fiber has cardinality 1 (or 2, the enantiomer pair).
- (P2) *Local identifiability.* The Fisher information operator $I_X = (DF_X)^*(DF_X)$ on $T_X C$ has full rank for all $X \in C$ (no flat directions; weighted by the scattering weights w_{ij}).
- (P3) *ε -convexity.* $\mathcal{L}_y|_C$ is geodesically ε -convex on the connected component of C containing $X_\star$ and has a unique stationary point there (no spurious local minima; distinct basins separated by barriers of height $> \varepsilon$).

Problem 1. *Characterize the chemistry-consistent priors C , as a function of the resolution $(Q_{\max}, \sigma)$ — equivalently of m_{eff} — for which the three properties above are equivalent and hold for generic $X_\star \in C$, and determine the minimal such prior.*

Concretely:

- (a) *Threshold.* Prove or disprove that there is a sharp threshold at $m_{\text{eff}} = d_C$: ε -convexification (and the equivalence (P1) $\Leftrightarrow$ (P2) $\Leftrightarrow$ (P3)) holds for generic C iff $m_{\text{eff}}(Q_{\max}, \sigma) \geq d_C$, subject to the chirality lower bound. Construct counterexamples at $m_{\text{eff}} = d_C - 1$ in which flips or spurious local minima persist.
- (b) *Minimal prior.* For given $(Q_{\max}, \sigma)$, find the maximal admissible d_C — the *least* chemical knowledge (most free dihedrals, fewest fixed bonds/angles/coordination constraints) — for which the threshold is met, and describe the trade-off curve $d_C^*(Q_{\max})$. (Low resolution forces small d_C^* , i.e. more chemistry; as $Q_{\max} \rightarrow \infty$, $d_C^* \uparrow$ and the rigidity limit is recovered.)
- (c) *Information-theoretic form.* Show that the threshold $m_{\text{eff}} \geq d_C$ coincides with the vanishing of the information deficit

$$I_{\text{deficit}} = h(X | C) - I(F(X); X | C),$$

up to the irreducible chirality term $\log 2$, where h is the differential entropy of the prior manifold and I the mutual information of the band-limited measurement.

The crux is that this is generic-global-rigidity theory transferred from *exact, unweighted, complete* distance data to *band-limited, scattering-weighted, soft* data on a *chemistry-constrained manifold*: the regime left open at the intersection of [1, 2, 3].

References

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